

Semantic Book Recommender

Technical Documentation & Architecture Guide

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1. Introduction & System Overview

1.1 Project Purpose

The Semantic Book Recommender is an advanced Natural Language Processing (NLP) application designed to transcend traditional metadata-based book searches. Instead of relying purely on exact keyword matches, author names, or static genres, this system utilizes deep learning models to understand the *semantic meaning* of a user's query and matches it against the narrative descriptions, inherent emotions, and categorized themes of thousands of books.

1.2 System Architecture

The repository consists of several core components that work in tandem to ingest data, extract meaningful features, embed text into a vector space, and serve recommendations via a user-friendly interface. The primary modules include:

- **Data Preprocessing:** Cleaning and standardizing raw book metadata (`books_cleaned.csv`).
- **Semantic Representation:** Converting textual data into high-dimensional vectors for similarity matching (`vector_search.ipynb`).
- **Feature Engineering (Classification):** Assigning genres and themes using zero-shot or fine-tuned text classifiers (`text-classification.ipynb`, `books_with_categories.csv`).
- **Feature Engineering (Sentiment & Emotion):** Extracting emotional arcs and sentiment scores from book descriptions (`sentiment-analysis.ipynb`, `books_with_emotions.csv`).
- **Interactive Dashboard:** A Gradio-based web application that allows users to input natural language queries and receive instant book recommendations (`gradio-dashboard.py`, `app.py`).

1.3 Environment Setup

To replicate this environment, a comprehensive `requirements.txt` is provided. The stack heavily relies on modern Python data science libraries.

```
1 # Create a virtual environment
2 python -m venv venv
3 source venv/bin/activate # On Windows use: venv\Scripts\activate
4
5 # Install dependencies
6 pip install -r requirements.txt
```

Basic Environment Setup

1.4 Live Deployments

The application has been deployed and is publicly accessible for demonstration and API testing. You can explore the live environments using the following links:

- **Frontend Application (Vercel):** <https://semantic-book-search.vercel.app/>
- **Backend API Documentation (Hugging Face Space):** <https://ahmed-ayman-book-recommender-backend.hf.space/docs>
- **Interactive Gradio Demo (Hugging Face Space):** <https://huggingface.co/spaces/ahmed-ayman/book-recommender>

2. Data Exploration & Preprocessing

2.1 Data Acquisition and Loading

The foundation of the pipeline is a robust dataset. The initial step involves fetching the "7k-books-with-metadata" dataset directly via Kaggle's hub and loading it into a pandas DataFrame for inspection.

```
1 import kagglehub
2 import pandas as pd
3
4 # Download latest version of the dataset
5 path = kagglehub.dataset_download("dylanjcastillo/7k-books-with-metadata")
6 print("Path to dataset files:", path)
7
8 # Load into DataFrame
9 books = pd.read_csv(path + "/books.csv")
```

Downloading and Loading Data

2.2 Missing Data & Correlation Analysis

Understanding the structure and completeness of the data is crucial. We visualize missing values across all columns and calculate the Spearman correlation between key numerical features, accounting for missing descriptions.

```
1 import seaborn as sns
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 # Visualize missing values
6 ax = plt.axes()
7 sns.heatmap(books.isna().transpose(), cbar=False, ax=ax)
8 plt.xlabel('columns')
9 plt.ylabel('missing')
10 plt.show()
11
12 # Engineer basic features for correlation
13 books['missing_description'] = np.where(books['description'].isna(), 1, 0)
14 books['age_of_book'] = 2025 - books['published_year']
15 columns_of_interest = ['num_pages', 'age_of_book', 'missing_description', '
    average_rating']
16
17 # Spearman correlation for non-continuous data (due to missing values)
18 correlation_matrix = books[columns_of_interest].corr(method='spearman')
19
20 sns.set_theme(style="white")
21 plt.figure(figsize=(8, 6))
22 heatmap = sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f",
    cbar_kws={'label': 'Spearman Correlation'})
23 heatmap.set_title('Correlation Heatmap', fontdict={'fontsize':12}, pad=12)
24 plt.show()
```

Visualizing Missing Values and Correlations

2.3 Data Filtering and Cleaning

To ensure high-quality recommendations, we must filter out records with missing core attributes (description, published year, average rating, number of pages). Furthermore, books with exceptionally short descriptions (e.g., "No Marketing Blurb" or "Fantasy-roman.") do not provide enough semantic context for our NLP models, so we enforce a minimum description length of 25 words.

```

1 # Drop rows with missing crucial data
2 book_missing = books[~(books["description"].isna()) &
3                       ~(books["published_year"].isna()) &
4                       ~(books["average_rating"].isna()) &
5                       ~(books["num_pages"].isna()))
6
7 # Calculate description length
8 book_missing["words_in_description"] = book_missing["description"].apply(lambda x: len(x)
9                                   .split())
10
11 # Filter out books with descriptions shorter than 25 words
12 book_missing_25_words = book_missing[book_missing["words_in_description"] >= 25]

```

Filtering Incomplete and Short Descriptions

2.4 Feature Engineering and Final Export

Finally, we consolidate relevant text fields. Titles and subtitles are combined to form a complete title string, and the ISBN is prepended to the description to create a unique tagged_description. Unnecessary columns are then dropped before exporting the final cleaned dataset.

```

1 # Combine Title and Subtitle
2 book_missing_25_words["title_and_subtitle"] = np.where(
3     book_missing_25_words["subtitle"].isna(),
4     book_missing_25_words["title"],
5     book_missing_25_words[["title", "subtitle"]].astype(str).agg(": ".join, axis=1)
6 )
7
8 # Create tagged description combining ISBN and Description
9 book_missing_25_words["tagged_description"] = book_missing_25_words[["isbn13", "
10     description"]].astype(str).agg(" ".join, axis=1)
11
12 # Drop intermediary columns and save
13 (
14     book_missing_25_words
15     .drop(["subtitle", "missing_description", "age_of_book", "words_in_description"],
16           axis=1)
17     .to_csv("books_cleaned.csv", index=False)
18 )
19 print("Cleaned dataset successfully saved.")

```

Feature Consolidation and Export

This rigorously cleaned dataset forms the basis for all subsequent NLP tasks.

3. Semantic Vector Search

3.1 LangChain Setup and Document Preparation

The `vector_search.ipynb` notebook demonstrates the semantic core of the application using LangChain. First, we extract the `tagged_description` column (which combines the 13-digit ISBN and the book's description) into a raw text file so it can be easily ingested by LangChain's `TextLoader`.

```
1 import pandas as pd
2 from langchain_community.document_loaders import TextLoader
3
4 # Load the cleaned dataset
5 books = pd.read_csv("books_cleaned.csv")
6
7 # Export tagged descriptions to a text file
8 books["tagged_description"].to_csv(
9     "tagged_description.txt",
10     sep="\n",
11     index=False,
12     header=False
13 )
14
15 # Load the documents using LangChain
16 raw_documents = TextLoader("tagged_description.txt", encoding="utf-8").load()
```

Exporting and Loading Text Documents

3.2 Text Chunking and Chroma Vector Store

To process the descriptions effectively, we chunk the text using a `RecursiveCharacterTextSplitter`. We then use a local open-source embedding model (`sentence-transformers/all-MiniLM-L6-v2`) via Hugging Face to convert these chunks into dense vector embeddings, which are stored in a Chroma vector database.

```
1 from langchain_text_splitters import RecursiveCharacterTextSplitter
2 from langchain_huggingface import HuggingFaceEmbeddings
3 from langchain_chroma import Chroma
4 import warnings
5 warnings.filterwarnings("ignore")
6
7 # Split text into meaningful chunks
8 text_splitter = RecursiveCharacterTextSplitter(
9     chunk_size=1000,
10    chunk_overlap=100,
11    separators=["\n\n", "\n", " ", ""]
12 )
13 documents = text_splitter.split_documents(raw_documents)
14
15 # Build Chroma Vector Database using local Hugging Face embeddings
16 db_books = Chroma.from_documents(
17     documents,
18     embedding=HuggingFaceEmbeddings(model_name="sentence-transformers/all-MiniLM-L6-v2")
19 )
```

Chunking and Embedding with Chroma

3.3 Mathematical Foundation: Cosine Similarity

At the core of our retrieval system is the ability to quantify the semantic closeness between a user's query and the available book descriptions. Once the `HuggingFaceEmbeddings` model converts both the query and the documents into high-dimensional dense vectors, we evaluate their proximity.

While vector databases can use various distance metrics (like L2 Euclidean distance), the semantic relationship is fundamentally understood through **Cosine Similarity**.

Cosine similarity measures the cosine of the angle between two non-zero vectors, A (the user query vector) and B (the document chunk vector), in a multi-dimensional embedding space. It is defined mathematically as:

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}} \quad (3.1)$$

Where:

- $A \cdot B$ is the dot product of the two vectors.
- $\|A\|$ and $\|B\|$ are the Euclidean norms (magnitudes) of the vectors.
- n is the number of dimensions in the embedding space (e.g., 384 dimensions for the all-MiniLM-L6-v2 model).

By normalizing the dot product by the vectors' magnitudes, cosine similarity focuses entirely on the *orientation* (semantic alignment) rather than the *magnitude* (length of the text). A resulting score of 1 indicates exact semantic alignment, 0 indicates orthogonality (completely unrelated concepts), and -1 indicates opposite meanings. Our Chroma vector store leverages this geometric relationship under the hood to rank and retrieve the top- k nearest neighbors (most relevant books) to the user's input.

3.4 Semantic Retrieval Function

When a user submits a natural language query (e.g., "A book to teach children about nature"), we query the Chroma vector store. Because each document chunk begins with the book's 13-digit ISBN, we use a regular expression to extract the ISBN from the retrieved documents and use it to look up the full book metadata in our Pandas DataFrame.

```
1 import re
2
3 def retrieve_semantic_recommendations(query: str, top_k: int = 10) -> pd.DataFrame:
4     # Search the Chroma database for the most similar chunks
5     recs = db_books.similarity_search(query, k=50)
6
7     books_list = []
8
9     # Extract ISBNs from the returned documents
10    for rec in recs:
11        # Use regex to find the 13-digit ISBN at the beginning of the string
12        match = re.match(r'^\d{13}', rec.page_content.strip(''))
13        if match:
14            isbn_str = match.group(0)
15            books_list.append(int(isbn_str))
16
17    # Filter the original dataframe to return matching books
18    return books[books["isbn13"].isin(books_list)].head(top_k)
19
20 # Example usage:
21 # retrieve_semantic_recommendations("A book to teach children about nature")
```

Retrieving Recommendations via Vector Similarity

4. Text Classification & Categorization

4.1 Methodology and Category Mapping

Traditional book categorization relies on publishers assigning varied and sometimes overly granular genres. Our approach, detailed in `text-classification.ipynb`, first normalizes the existing categories into a simplified taxonomy (e.g., Fiction, Nonfiction, Children's Fiction, Children's Nonfiction).

```
1 import pandas as pd
2
3 # Load cleaned data
4 books = pd.read_csv("books_cleaned.csv")
5
6 # Map detailed categories to a simplified taxonomy
7 category_mapping = {
8     'Fiction' : "Fiction",
9     'Juvenile Fiction': "Children's Fiction",
10    'Biography & Autobiography': "Nonfiction",
11    'History': "Nonfiction",
12    'Literary Criticism': "Nonfiction",
13    'Philosophy': "Nonfiction",
14    'Religion': "Nonfiction",
15    'Comics & Graphic Novels': "Fiction",
16    'Drama': "Fiction",
17    'Juvenile Nonfiction': "Children's Nonfiction",
18    'Science': "Nonfiction",
19    'Poetry': "Fiction"
20 }
21
22 books["simple_categories"] = books["categories"].map(category_mapping)
```

Category Simplification Mapping

4.2 Zero-Shot Classification for Missing Categories

For books missing these simplified categories, we utilize transformer-based models to dynamically assign them. By leveraging Hugging Face's pipeline with the `facebook/bart-large-mnli` model, we can perform zero-shot classification to predict whether a book is "Fiction" or "Nonfiction" based entirely on its description.

```
1 from transformers import pipeline
2 import torch
3 import numpy as np
4
5 # Set device to GPU if available, else CPU
6 device = 0 if torch.cuda.is_available() else -1
7
8 # Initialize zero-shot classification pipeline
9 pipe = pipeline("zero-shot-classification",
10                 model="facebook/bart-large-mnli",
11                 device=device)
12
13 fiction_categories = ["Fiction", "Nonfiction"]
14
15 def generate_predictions(sequence, categories):
16     predictions = pipe(sequence, categories)
17     max_index = np.argmax(predictions["scores"])
18     return predictions["labels"][max_index]
```

Zero-Shot Text Classification Setup

4.3 Imputation and Finalizing Categories

After testing the model's accuracy on a known subset (achieving approximately 78% accuracy), we iterate over the records containing missing `simple_categories` to impute the predicted values.

```
1  from tqdm import tqdm
2
3  isbnns = []
4  predicted_cats = []
5
6  # Isolate books with missing simple categories
7  missing_cats = books.loc[books["simple_categories"].isna(), ["isbn13", "description"]].
    reset_index(drop=True)
8
9  # Generate predictions for missing data
10 for i in tqdm(range(0, len(missing_cats))):
11     sequence = missing_cats["description"][i]
12     predicted_cats.append(generate_predictions(sequence, fiction_categories))
13     isbnns.append(missing_cats["isbn13"][i])
14
15 # Merge predictions back into the main DataFrame
16 missing_predicted_df = pd.DataFrame({"isbn13": isbnns, "predicted_categories":
    predicted_cats})
17 books = pd.merge(books, missing_predicted_df, on="isbn13", how="left")
18
19 # Fill missing simple_categories with the predictions
20 books["simple_categories"] = np.where(
21     books["simple_categories"].isna(),
22     books["predicted_categories"],
23     books["simple_categories"]
24 )
25 books = books.drop(columns=["predicted_categories"])
26
27 # Save the final dataset
28 books.to_csv("books_with_categories.csv", index=False)
```

Imputing Missing Categories

The output of this process is saved to `books_with_categories.csv`, enriching the dataset with a complete, baseline categorization suitable for downstream recommendation filtering.

5. Sentiment & Emotion Analysis

5.1 Extracting Emotional Arcs

A unique feature of this recommender system is its ability to understand the “vibe” or emotional undertone of a book. A user might want a “dark, suspenseful thriller” or an “uplifting, joyful romance.” The `sentiment-analysis.ipynb` notebook generates the `books_with_emotions.csv` dataset by analyzing the narrative descriptions.

5.2 Sentence-Level Emotion Pooling

Instead of evaluating an entire description at once—which can cause strong emotional moments to be diluted by neutral exposition—the pipeline splits each description into individual sentences. We then run the `j-hartmann/emotion-english-distilroberta-base` model on every sentence and extract the *maximum* score achieved for each of the seven core emotions (Anger, Disgust, Fear, Joy, Neutral, Sadness, Surprise). This ensures the peak emotional themes of the book are captured.

```
1 import pandas as pd
2 import numpy as np
3 from transformers import pipeline
4
5 # Load data with categorized features
6 books = pd.read_csv("books_with_categories.csv")
7
8 # Initialize text classification pipeline (using MPS/GPU for speed)
9 classifier = pipeline("text-classification",
10                       model="j-hartmann/emotion-english-distilroberta-base",
11                       top_k=None,
12                       device="mps")
13
14 emotion_labels = ["anger", "disgust", "fear", "joy", "sadness", "surprise", "neutral"]
15
16 def calculate_max_emotion_scores(predictions):
17     """
18     Takes predictions for multiple sentences and returns the maximum
19     score achieved for each emotion across the entire text.
20     """
21     per_emotion_scores = {label: [] for label in emotion_labels}
22     for prediction in predictions:
23         sorted_predictions = sorted(prediction, key=lambda x: x["label"])
24         for index, label in enumerate(emotion_labels):
25             per_emotion_scores[label].append(sorted_predictions[index]["score"])
26
27     return {label: np.max(scores) for label, scores in per_emotion_scores.items()}
```

Emotion Pipeline and Max-Pooling Setup

5.3 Processing the Dataset

With the max-pooling function defined, we iterate over the entire catalog, parsing the descriptions sentence-by-sentence. The resulting peak emotion scores are merged back into the main dataset.

```
1 from tqdm import tqdm
2
3 isbn = []
4 emotion_scores = {label: [] for label in emotion_labels}
5
6 # Iterate through every book description
7 for i in tqdm(range(len(books))):
8     isbn.append(books["isbn13"][i])
9
```

```

10     # Split description into sentences
11     sentences = books["description"][i].split(".")
12
13     # Predict emotions for each sentence
14     predictions = classifier(sentences)
15
16     # Extract peak emotion scores
17     max_scores = calculate_max_emotion_scores(predictions)
18     for label in emotion_labels:
19         emotion_scores[label].append(max_scores[label])
20
21 # Construct the DataFrame of emotions
22 emotions_df = pd.DataFrame(emotion_scores)
23 emotions_df["isbn13"] = isbn
24
25 # Merge with the main dataset and export
26 books = pd.merge(books, emotions_df, on="isbn13")
27 books.to_csv("books_with_emotions.csv", index=False)

```

Extracting and Merging Emotion Scores

These continuous emotional scores allow for sophisticated vector math. If a user asks for a "happy" book, we can bias the semantic search results towards items with a high "joy" score.

6. User Interface: Gradio Dashboard

6.1 Overview

The final layer of the application is a user-friendly web interface built with Gradio. This allows non-technical users to interact with the complex ML backend simply by typing what they are in the mood to read, and further refining their search using category and emotional tone filters derived from our earlier data engineering steps.

6.2 Application Implementation

The application loads the `books_with_emotions.csv` dataset and dynamically initializes the Chroma vector store using the `tagged_description.txt` file. It incorporates advanced filtering logic: after retrieving the top semantic matches, it filters the results by the user's selected category and sorts them by the selected emotional tone (e.g., Joy for "Happy", Fear for "Suspenseful").

```
1 # =====
2 # Imports & Setup
3 # =====
4 import os
5 import numpy as np
6 import pandas as pd
7 if __name__ == "__main__":
8     dashboard.launch()
```

Gradio Application Dashboard

6.3 Deployment Considerations

For deployment, tools like Docker or platforms like Hugging Face Spaces can be utilized. The `requirements.txt` must accurately reflect versions of `torch`, `transformers`, `langchain`, `langchain-chroma`, `pandas`, and `gradio`. Because Chroma generates an in-memory or file-based database upon startup in this script, deployments on ephemeral instances (like standard Docker containers) will experience a slight delay on the very first run as it ingests and chunks the `tagged_description.txt` file.

7. Modernized Full-Stack Architecture & CI/CD

7.1 Evolution to a Decoupled System

While the initial iteration of the project utilized Gradio for rapid UI prototyping, the production deployment has been refactored into a modern, decoupled full-stack architecture. This structure clearly separates the Python machine learning backend from the user-facing web frontend, significantly improving scalability, maintainability, and deployment flexibility.

7.2 Backend REST API

The backend service, housed in the `backend/` directory, serves as the computational engine. It encapsulates the Chroma vector database and exposes the recommendation logic via a secure REST API (such as FastAPI or Flask), which allows any client to request semantic matches.

```
1  from fastapi import FastAPI
2  from pydantic import BaseModel
3
4  app = FastAPI(title="Semantic Book Recommender API")
5
6  class QueryRequest(BaseModel):
7      query: str
8      category: str = "All"
9      tone: str = "All"
10
11  @app.post("/api/recommend")
12  def get_recommendations(req: QueryRequest):
13      # Calls the core LangChain/Chroma retrieval function
14      recs = retrieve_semantic_recommendations(
15          query=req.query,
16          category=req.category,
17          tone=req.tone
18      )
19      # Return matched records as a JSON list
20      return {"recommendations": recs.to_dict(orient="records")}
```

Backend API Route Example (backend/main.py)

7.3 Next.js Frontend Application

The client interface, located in the `frontend/` directory, is built using the Next.js framework and React. This responsive frontend fetches and displays book recommendations dynamically by calling the Python backend's exposed endpoints over HTTP.

```
1  "use client";
2  import { useState } from 'react';
3
4  export default function BookRecommender() {
5      const [query, setQuery] = useState('');
6      const [books, setBooks] = useState([]);
7
8      const handleSearch = async () => {
9          // Ping the backend API for semantic recommendations
10         const response = await fetch('https://api.example.com/api/recommend', {
11             method: 'POST',
12             headers: { 'Content-Type': 'application/json' },
13             body: JSON.stringify({ query, category: 'All', tone: 'All' })
14         });
15         const data = await response.json();
```

```

16     setBooks(data.recommendations);
17   };
18
19   return (
20     <main className="p-8">
21       <input
22         className="border p-2 rounded"
23         value={query}
24         onChange={(e) => setQuery(e.target.value)}
25         placeholder="What do you want to read?"
26       />
27       <button onClick={handleSearch} className="bg-blue-500 p-2 text-white ml-2 rounded"
28       >
29         Search
30       </button>
31       { /* Map and render book results here */ }
32     </main>
33   );
34 }

```

Frontend Data Fetching (frontend/app/page.tsx)

7.4 Continuous Integration with GitHub Actions

To maintain high availability for the deployed backend (hosted on Hugging Face Spaces), the repository includes a GitHub Actions workflow. Free tier Spaces spin down after periods of inactivity, so the `keepalive.yml` script periodically pings the server to keep it awake and continuously responsive for the frontend application.

```

1 name: Keep HuggingFace Space Awake
2
3 on:
4   schedule:
5     # Run automatically every 12 hours
6     - cron: '0 */12 * * *'
7   workflow_dispatch:
8
9 jobs:
10  ping:
11    runs-on: ubuntu-latest
12    steps:
13      - name: Ping Backend API
14        run: |
15          curl -s https://ahmed-ayman-book-recommender-backend.hf.space
16          echo "Space successfully pinged and kept awake!"

```

Keepalive Workflow (.github/workflows/keepalive.yml)

8. Conclusion and Future Enhancements

8.1 Summary

The Semantic Book Recommender represents a modern, NLP-driven approach to discovery. By moving away from rigid keyword indexing and embracing dense vector representations orchestrated through LangChain and a Chroma vector database, we capture the profound nuances of human language. Incorporating zero-shot categorization and sentence-level peak emotion detection provides a rich, multifaceted dataset that powers complex, highly personalized search queries through an intuitive Gradio interface.

8.2 Future Work

While the current architecture is robust, the modular design provided by LangChain allows for several advanced enhancements in future iterations:

1. **Retrieval-Augmented Generation (RAG):** Leveraging the existing LangChain and Chroma architecture to introduce a conversational Large Language Model (such as Llama 3 or Gemini). This would transform the system from a static recommender into an interactive agent capable of explaining *why* a specific book fits the user's prompt based on the retrieved context chunks.
2. **Ensemble Retrieval (Hybrid Search):** Combining the dense semantic vector search currently implemented in Chroma with a sparse lexical search algorithm (like BM25). By utilizing LangChain's `EnsembleRetriever`, the system could elegantly balance semantic meaning with exact title or author keyword matches.
3. **User Personalization:** Implementing collaborative filtering or a user-profile embedding mechanism. If a user logs in and rates books, their historical preferences could be vectorized and averaged with their current search query vector to highly personalize the output.
4. **Dynamic Data Pipelines:** Implementing an Airflow or Prefect pipeline to routinely scrape new book releases. Using LangChain's document loaders, the pipeline could clean the data, run inferences for categories and emotions, and incrementally update the Chroma vector store without downtime or full re-indexing.

End of Documentation.